

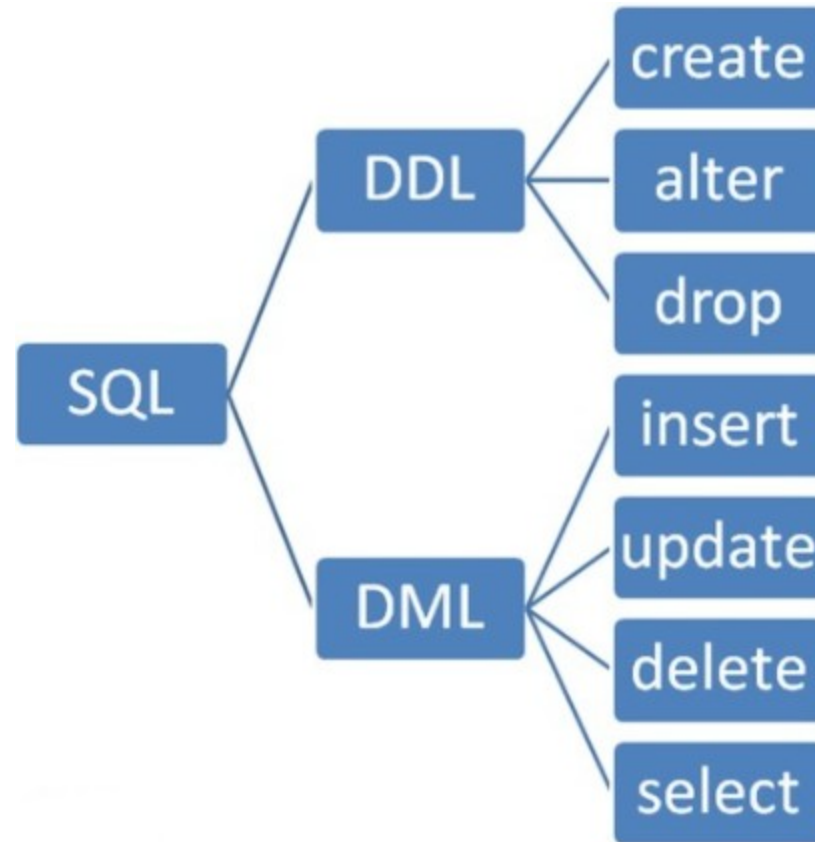
Basic SQL

(Tool: MYSQL Workbench)

SQL

Data Definition Language

Data Manipulation Language



Main Commands for DDL

Using DDL commands we can make changes in the structure of the database.

Main Commands:

- Create – used to create a new table in the database
- Alter – used to make changes in the already created tables
- Drop – used to delete an existing table

Getting Started ...

Before creating a table in the database, we need to create a database itself:

- Create database `databaseName`;
- Example:
 - Create database Company;
- SQL is not case sensitive.
- Every command needs to end with a semicolon.

Getting Started ...

We can delete a database :

Drop database DatabaseName;

Example:

Drop database Company;

Getting Started ...

Before creating a new table or viewing already constructed tables we need to make sure we are in scope of the database or we are present in a database.

Use `databaseName` ;

Example:

Use `Company`;

We can see tables present in an existing database:

Example:

Show `tables`;

How to create a table

Datatypes & Constraints

Datatypes

Numeric values

INT (Size: 4 Bytes)

NUMERIC(I, j) 888.88

Fixed point number, with user-specified precision of p digits, with d digits to the right of decimal point

Character String

Fixed length: CHAR(5)
CAT__

Non-fixed length:
VARCHAR(n) CAT

Some more important datatypes

Boolean: TRUE, FALSE, NULL

DATE : yyyy-mm-dd

TIME: hh:mm:ss

TIMESTAMP: DATE(DD-MM-YYYY) & TIME (HH:MM:SS)

Key words for Constraints

NOT NULL – used for required fields

UNIQUE – used for attributes that cannot have a duplicate value (other than pk)

DEFAULT – used to assign some default value to an attribute

CHECK(Dnumber > 0 AND Dnumber < 21) -- used to set a condition on the input value for an attribute.

Primary key – used to make an attribute primary key of the table

Foreign Key (A) references s(B) -- used to make an attribute foreign key.

Create Table

Important Notes

- 1 Make sure a database is selected

CREATE TABLE tablename

(

AttributeName Datatype Constraint,

.....

);

Safe side of creation

CREATE TABLE IF NOT EXISTS students (id INT PRIMARY KEY, name VARCHAR(50));

Example (Full Flow)

```
CREATE DATABASE IF NOT EXISTS my_database;
```

```
USE my_database;
```

```
CREATE TABLE IF NOT EXISTS students ( id INT PRIMARY KEY, name VARCHAR(50)); (for test delete after this using  
drop)  
drop table students;
```

```
CREATE TABLE departments (dept_id INT PRIMARY KEY,dept_name VARCHAR(50) NOT NULL);  
CREATE TABLE students (student_id INT PRIMARY KEY, name VARCHAR(50) NOT NULL,dept_id INT,FOREIGN KEY  
(dept_id) REFERENCES departments(dept_id));
```

Optional(constraint for foreign key)

```
CREATE TABLE students ( student_id INT PRIMARY KEY, name VARCHAR(50) NOT NULL, dept_id INT,  
FOREIGN KEY (dept_id) REFERENCES departments(dept_id) ON DELETE CASCADE ON UPDATE CASCADE);
```

Employee

CREATE TABLE Employee

(

Fname,

Minit,

Lname,

Ssn,

Bdat,

Address,

Gender,

Salary,

Super_ssn ,

Dno ,

CONSTRAINT PK_Employee PRIMARY KEY (attributes),

CONSTRAINT FK_EmpDep FOREIGN KEY (attribute)

REFERENCES table name(attribute));

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	gen	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

<u>Easn</u>	<u>Pno</u>	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

<u>Easn</u>	<u>Dependent_name</u>	gen	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Employee

CREATE TABLE Employee

```
(  
  Fname VARCHAR(15) NOT NULL,  
  Minit CHAR(1),  
  Lname VARCHAR(15) NOT NULL,  
  Ssn CHAR (9) NOT NULL,  
  Bdat DATE,  
  Address VARCHAR(30) DEFAULT 'UCP',  
  Gender Char(1),  
  Salary int,  
  Super_ssn CHAR(9),  
  Dno INT ,  
  Primary key (ssn),  
  CONSTRAINT FK_EmpDep FOREIGN KEY (Dno)  
    REFERENCES DEPARTMENT(Dnumber) );
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	gen	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
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Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
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James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

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987987987	30	5.0
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987654321	20	15.0
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PROJECT

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DEPENDENT

Easn	Dependent_name	gen	Bdate	Relationship
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333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Solution of Foreign key violation

We can specify RESTRICT, CASCADE, SET NULL or SET DEFAULT on referential integrity constraints (foreign keys)

- On delete / Update **Cascade / Restrict**

Automatically delete / update data

- On delete / update **Set NULL / Default**

Automatically set Null.

On delete no action is performed by default

EMPLOYEE

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Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
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Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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5	Houston

WORKS_ON

<u>Easn</u>	<u>Pno</u>	Hours
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333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

<u>Easn</u>	<u>Dependent_name</u>	gen	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Employee

CREATE TABLE Employee

(

Fname VARCHAR(15) NOT NULL,

Minit CHAR(1),

Lname VARCHAR(15) NOT NULL,

Ssn CHAR (9) NOT NULL,

Bdat DATE,

Address VARCHAR(30) DEFAULT 'UCP',

Gender Char(1),

Salary int,

Super_ssn CHAR(9),

Dno INT DEFAULT 1,

Primary key (ssn),

CONSTRAINT FK_EmpDep FOREIGN KEY (Dno)

REFERENCES DEPARTMENT(Dnumber) ON DELETE SET
DEFAULT ON UPDATE CASCADE

);

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	gen	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
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Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

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WORKS_ON

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123456789	2	7.5
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333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
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999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Easn	Dependent_name	gen	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Department

```
CREATE TABLE DEPTARTMENT
( Dname VARCHAR(15) NOT NULL ,
  Dnumber INT CHECK (Dnumber > 0 AND Dnumber<21),
  Mgr_ssn CHAR(9),
  Mgr_start_date DATE,

  PRIMARY KEY (Dnumber),
  UNIQUE (Dname),
  Constraint DepMgrFK FOREIGN KEY (Mgr_ssn) REFERENCES
Employee(Ssn) ON DELETE SET NULL ON UPDATE CASCADE
);
```

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	gen	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

<u>Esn</u>	<u>Pno</u>	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

<u>Esn</u>	<u>Dependent_name</u>	gen	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

Works_on

CREATE TABLE WORKS_ON

(

Essn CHAR(9),

Pno INT,

Hours DECIMAL (3,1),

Primary key(Essn, Pno),

Constraint essnfk Foreign key (ESSN) References EMPLOYEE (ssn)

ON DELETE SET NULL ON UPDATE CASCADE

);

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	gen	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
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Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
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DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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WORKS_ON

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123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
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Class Activity

Create table of
DEPT_LOCATION
PROJECT
DEPENDENT

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	gen	Salary	Super_ssn	Dno
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999887777	10	10.0
987987987	10	35.0
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Newbenefits	30	Stafford	4

DEPENDENT

<u>Easn</u>	<u>Dependent_name</u>	gen	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

DROP TABLE

Used to remove a relation (table) *and its definition*

The relation can no longer be used in queries, updates, or any other commands since its description no longer exists

Example:

DROP TABLE DEPENDENT;

ALTER TABLE

Used to add an attribute to one of the tables

Used to drop an existing attribute of the table

Used to change the datatype of an existing attribute of the table

Used to change the name of an existing attribute of the table

Used to add/drop constraints

-
- Add column - Adds a new column to a table
 - Drop column - Deletes a column in a table
 - Rename column - Renames a column
 - Modify column - Changes the data type, size, or constraints of a column
 - Add constraint - Adds a new constraint
 - Rename table - Renames a table

Add/Drop attribute

The new attribute will have NULLs in all the tuples of the relation right after the command is executed; hence, the NOT NULL constraint is *not allowed* for such an attribute

Example:

```
ALTER TABLE EMPLOYEE ADD JOB VARCHAR(12);
```

```
ALTER TABLE EMPLOYEE DROP JOB ;
```

Setting position for a new attribute

By default attribute is added at the end of the table

student

<u>Sid</u>	Name	Address
------------	------	---------

```
ALTER TABLE student ADD Email varchar(50) FIRST;
```

```
ALTER TABLE student DROP Email;
```

```
ALTER TABLE student ADD Email varchar(50) AFTER Sid;
```

Changing a Column Datatype or Name

ALTER TABLE student Modify Email varchar(100);

student

<u>Sid</u>	Name	Address
------------	------	---------

ALTER TABLE student Change Email Email_Address varchar(100);

Change command can change both datatype and name.

ALTER TABLE student Change Email Email_Address varchar(50);

Adding Not Null and Default Value of Attributes

student

<u>Sid</u>	Name	Address
------------	------	---------

```
ALTER TABLE student MODIFY Address Text NOT NULL DEFAULT 'LAHORE';
```

TO change Default value:

```
ALTER TABLE student ALTER Address SET DEFAULT 'KARACHI';
```

You can remove default constraint from any column:

```
ALTER TABLE student ALTER Address drop default;
```

Rename a table

```
ALTER TABLE Student RENAME TO UnderGrad_student;
```

Drop/ add primary key

Alter table employee

Drop primary key;

Alter table employee

Add primary key (ssn);

Drop/add foreign key

Alter table employee

Add constraint fk_1 foreign key (dno) references department (dnumber);

Alter table employee

Drop foreign key fk_1;